Supplementary Materials for A Formal Investigation of Paradoxes Between Latent Aptitude and Observed Achievement via Full House Methodology

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1 Monotonicity

Lemma 1.1. *Let $1 \leq j < n$ and let $x \in (0,1)$. Let $I_{\alpha,\beta}(x)$ be a regularized incomplete beta function as in Definition 3.3. Then, $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$.*

Proof. We can define a function $f_{n,j}(x)$ as

$$\begin{aligned} f_{n,j}(x) &= I_{n-j,j+1}(x^{\frac{1}{n}}) - I_{n-j+1,j+1}(x^{\frac{1}{n+1}}) \\ &= \int_0^{x^{\frac{1}{n}}} \frac{y^{n-j-1}(1-y)^j}{B(n-j,j+1)}dy - \int_0^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1,j+1)}dy. \end{aligned}$$

An application of the chain rule yields

$$\begin{aligned} f'_{n,j}(x) &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n+1)B(n-j+1,j+1)} \\ &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n-j)B(n-j,j+1)} \\ &= \frac{1}{B(n-j,j+1)}\left(\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}\right), \end{aligned}$$

where we used $B(n-j+1,j+1) = \frac{n-j}{n+1}B(n-j,j+1)$ in the second equality. Now define $A_{n,j} = \{x : f'_{n,j}(x) \geq 0\}$. For $x \in A_{n,j}$ we have

$$\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} \geq \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}$$

$$\implies \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^{j}\left(x^{\frac{1-n}{n}}\right)}{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j}\left(x^{-\frac{n}{n+1}}\right)} \geq \frac{n}{n-j}$$

$$\implies x^{-\frac{j}{n(n+1)}}\left(\frac{1-x^{\frac{1}{n}}}{1-x^{\frac{1}{n+1}}}\right)^{j} \geq \frac{n}{n-j}$$

$$\implies \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}} \geq \left(\frac{n}{n-j}\right)^{\frac{1}{j}}.$$

Define $g_n(x) = \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}$. Then

$$g_n(x) = \frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}} = \frac{x^{-\frac{1}{n(n+1)}}-1}{1-x^{\frac{1}{n+1}}}+1,$$

and

$$\begin{aligned} g_n^{'}(x) &= \frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(-\frac{1}{n(n+1)}x^{-\frac{1}{n(n+1)}-1}(1-x^{\frac{1}{n+1}}) - (x^{-\frac{1}{n(n+1)}}-1)(-\frac{1}{n+1}x^{\frac{1}{n+1}-1})\right) \\ &= \frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(\frac{1}{n}x^{-\frac{n^2+1}{n(n+1)}} - \frac{1}{n+1}x^{-\frac{n}{n+1}} - \frac{1}{n(n+1)}x^{-\frac{n^2+n+1}{n(n+1)}}\right) \\ &= \frac{x^{-\frac{n^2+1}{n(n+1)}}}{n(n+1)(1-x^{\frac{1}{n+1}})^2}\left((n+1) - nx^{\frac{1}{n(n+1)}} - x^{-\frac{1}{n+1}}\right). \end{aligned}$$

Now define $l_n(x) = (n+1) - nx^{\frac{1}{n(n+1)}} - x^{-\frac{1}{n+1}}$ and see that

$$l_n^{'}(x) = -\frac{1}{n+1}x^{\frac{1}{n(n+1)}-1} + \frac{1}{n+1}x^{-\frac{1}{n+1}-1} = \frac{1}{n+1}x^{-\frac{n+2}{n+1}}\left(1-x^{\frac{1}{n}}\right).$$

Thus, $l_n(x)$ is an increasing function for $x \in (0,1)$ and $\sup_{x\in(0,1)} l_n(x) = 0$. Therefore, $l_n(x) < 0$ for $x \in (0,1)$. Thus, $g_n(x)$ is an decreasing function, and $g_n(x) \to 1$ as $x \to 1$. We can see that $g_n(x) \in (1, +\infty)$ for $x \in (0,1)$. Thus $A_{n,j} = (0, x^*_{n,j}]$ where $x^*_{n,j}$ is such that $g_n(x^*_{n,j}) = (\frac{n}{n-j})^j$. Putting it all together, $f_{n,j}(0) = f_{n,j}(1) = 0$ and this implies that $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$ for all $x \in (0,1)$. □

2 Proof to Lemma: 4.4

Proof. Notice that the cumulative distribution function of multivariate normal distribution can be decomposed into a series of product of conditional univariate normal distribution.

$$\begin{aligned} \mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) = \Phi(W_n = w_n | W_1 = w_1, W_2 = w_2, \ldots, W_{n-1} = w_{n-1}) \times \\ \Phi(W_{n-1} = w_{n-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n-2} = w_{n-2}) \times \cdots \times \\ \Phi(W_2 = w_2 | W_1 = w_1) \times \Phi(W_1 = w_1), \end{aligned} \tag{1}$$

where $\mathbf{W} = (W_1, W_2, \ldots, W_n)$, $\mathbf{w} = (w_1, w_2, \ldots, w_n)$.

Then in general, for any $k \in \{2, \ldots, n_1 + n_2 - 1\}$, the conditional distribution of W_k given $W_1, \ldots, W_{k-1}$ is

$$W_k \mid W_1 = w_1, \ldots, W_{k-1} = w_{k-1} \sim \mathcal{N}\left(\mu_k, \sigma_k^2\right)$$

with

$$\mu_k = \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{\Sigma}_{1:(k-1),k}$$

Here: $\mathbf{\Sigma}_{k,1:(k-1)}$ is a $1 \times (k-1)$ row vector: the covariances between W_k and $W_1, \ldots, W_{k-1}$ $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ is a $(k-1) \times (k-1)$ covariance matrix, $\mathbf{\Sigma}_{k,k}$ is the kth element on the diagonal of the covariance matrix $\mathbf{\Sigma}$, and $\mathbf{w}_{1:(k-1)} = (w_1, \ldots, w_{k-1})^T$.

Then we will show the the parameters μ and σ^2 of the conditional univariate normal distribution in the set of Equation (16) are finite.

When $k \leq n_1$, by applying the Equation (15)

$$\begin{aligned}
\mu_k &= \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2 - \mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2 - \mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1) + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}(k-2) + 1 & -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2} \\ -\frac{\sigma_2^2}{\sigma_1^2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}(k-2) + 1 \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2 - \mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2 - \mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1) + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\mu_2 - \mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2 - \mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2^2(\mu_2 - \mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1) + 1}
\end{aligned}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)} \mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} \mathbf{\Sigma}_{1:(k-1),k}$$

$$
\begin{aligned}
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\sigma_2^2}{\sigma_1^2} + 1 - \frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

Then

$$
\begin{aligned}
&\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\frac{\sigma_2^2(\mu_2-\mu_1)}{\sigma_1^3}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{\frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(\frac{\frac{\mu_2-\mu_1}{\sigma_1}(1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}{\sqrt{1+\frac{\sigma_2^2}{\sigma_1^2}(1-\frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1})}} \right) \\
&= \Phi\left(\frac{\frac{\frac{\mu_2-\mu_1}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{1+\frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(\frac{\frac{\sigma_1(\mu_2-\mu_1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi(0) = 0.5
$$

where $\Phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

When $k = n_1 + 1$, the only difference from the case $k \leq n_1$ is that $\mathbf{\Sigma}_{k,\,1:(k-1)}$ is different. We can then apply a similar method as in the case $k \leq n_1$ to compute μ_k and σ_k.

$$
\begin{aligned}
\mu_k &= \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)} \\
&= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

$$
\begin{aligned}
\sigma_k^2 &= \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k} \\
&= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}^{-1}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-1)\times 1} \\
&= 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}
\end{aligned}
$$

Then

$$
\begin{aligned}
&\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \dots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\sigma_2(\mu_2-\mu_1)}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}{\sqrt{2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}(k-1)}{\frac{\sigma_2^2}{\sigma_1^2}(k-1)+1}}} \right)
\end{aligned}
$$

$$= \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right)$$
$$= c_k$$

$$\lim_{k\to+\infty} c_k = \lim_{k\to+\infty} \Phi\left(- \frac{\frac{\sigma_2(\mu_2-\mu_1)(k-1)}{\sigma_2^2(k-1)+\sigma_1^2}}{\sqrt{1+\frac{\sigma_2^2}{\sigma_2^2(k-1)+\sigma_1^2}}} \right) = \Phi\left(- \frac{\mu_2-\mu_1}{\sigma_2} \right)$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

When $k > n_1 + 1$, we first compute $\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}$. Based on the structure of $\mathbf{\Sigma}$ in Equation (10), the $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ can be decomposed into

$$\mathbf{\Sigma}_{1:(k-1),1:(k-1)} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \tag{2}$$

defined with the following structure

$$D_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{n_1\times n_1} \quad D_{12}^T = D_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(k-n_1-1)\times n_1} \tag{3}$$

$$D_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(k-n_1-1)\times(k-n_1-1)}.$$

By applying the Equation (15) and Schur complement [Zhang, 2006],

$$\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)} = \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}, \tag{4}$$

$$E_{11} = (D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}$$
$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \cdot \frac{1}{k-n_1} \cdot \begin{bmatrix} k-n_1-1 & -1 & \cdots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & k-n_1-1 \end{bmatrix} \right.$$

$$\left. \cdot \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right)^{-1} = \left($$

$$= \begin{bmatrix} \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} \\ \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \cdots & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2} & \frac{\sigma_2^2}{(k-n_1)\sigma_1^2}+1 \end{bmatrix}^{-1}$$

$$= \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}_{n_1 \times n_1}$$

$$E_{12} = -(D_{11} - D_{12}D_{22}^{-1}D_{21})^{-1}D_{12}D_{22}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$

$$\times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 2 \end{bmatrix}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$

$$\times \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{k-n_1} \begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix}$$

$$\times \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \dots & \dots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \dots & \dots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{n_1 \times (k-n_1-1)}$$

Since $\mathbf{\Sigma}_{1:(k-1),1:(k-1)}$ is a symmetric matrix, then $E_{12} = E_{21}^T$. Thus,

$$E_{21} = -D_{22}^{-1} D_{21} (D_{11} - D_{12} D_{22}^{-1} D_{21})^{-1}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1-1)\times n_1}$$

$$E_{22} = D_{22}^{-1} + D_{22}^{-1} D_{21} (D_{11} - D_{12} D_{22}^{-1} D_{21})^{-1} D_{12} D_{22}^{-1}$$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$= \frac{1}{k-n_1}\begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{(k-n_1)\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{(k-n_1)\sigma_1} \end{bmatrix}$$

$$= \frac{1}{k-n_1}\begin{bmatrix} k-n_1-1 & -1 & \dots & -1 \\ -1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \dots & -1 & k-n_1-1 \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}$$

$$= \begin{bmatrix} 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1-\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix}_{(k-n_1)\times(k-n_1)}$$

Then

$$\mu_k = \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{w}_{1:(k-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{(k-1)\times 1}$$

$$= \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1}+1 \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\mu_2-\mu_1}{\sigma_1} \\ \vdots \\ \frac{\mu_2-\mu_1}{\sigma_1} \end{bmatrix}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1(k-n_1-1)}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$= \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}$$

$$\sigma_k^2 = \mathbf{\Sigma}_{k,k} - \mathbf{\Sigma}_{k,1:(k-1)}\mathbf{\Sigma}^{-1}_{1:(k-1),1:(k-1)}\mathbf{\Sigma}_{1:(k-1),k}$$

$$= 2 - \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T_{(k-1)\times 1} \begin{bmatrix} E_{11} & E_{12} \\ E_{21} & E_{22} \end{bmatrix}_{(k-1)\times(k-1)} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{(k-1)\times 1}$$

$$= 2 - \left[\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \quad , \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \right] \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{11} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{21} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} E_{12} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} E_{22} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg)$$

$$= 2 - \Bigg(\begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1} + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1} + 1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{k-n_1} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{k-n_1} + 1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix}$$

$$+ \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} 1 - \frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \\ -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \dots & -\frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & 1 - \frac{\frac{1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \Bigg)$$

$$
\begin{aligned}
&= 2 - \left(\begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2}{\sigma_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} - \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2}{\sigma_1}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} \\ \vdots \\ \frac{\sigma_2}{\sigma_1} \end{bmatrix} \right. \\
&\left. - \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} 1 - \frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} & \cdots & 1 - \frac{\frac{k-n_1-1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right) \\
&= 2 - \left(\frac{\frac{\sigma_2^2}{\sigma_1^2}n_1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{2(k-n_1-1)n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} + \frac{k-n_1-1}{k-n_1} + \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{(k-n_1-1)^2 n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1} \right) \\
&= 2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}
\end{aligned}
$$

Then

$$
\begin{aligned}
&\Phi(W_k = w_k | W_1 = w_1, W_2 = w_2, \ldots, W_{k-1} = w_{k-1}) \\
&= \Phi\left(\frac{0 - \frac{\frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2}{\sigma_1}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2}\frac{n_1}{k-n_1}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{(k-n_1)^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{k-n_1}+1}}} \right) \\
&= \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}} \right) \\
&= c_k
\end{aligned}
$$

$$
\lim_{k \to +\infty} c_k = \lim_{k \to +\infty} \Phi\left(- \frac{\frac{(\mu_2-\mu_1)\sigma_2 n_1}{\sigma_2^2 n_1 + \sigma_1^2(k-n_1)}}{\sqrt{2 - \frac{k-n_1}{k-n_1+1} - \frac{\sigma_2^2 n_1}{\sigma_2^2 n_1(k-n_1) + \sigma_1^2(k-n_1)^2}}} \right) = \Phi(0) = 0.5
$$

where $\phi(x)$ is the cumulative distribution function of standard normal distribution.

Therefore, no matter what the value of k is, $c_k < 1$.

Therefore, we show the parameters μ and σ^2 of the conditional univariate normal distribution in the set of Equation (16) are finite.

Let

$$c_{max} = \max_{k=2,\ldots,n_1+n_2-1} c_k,$$

then $c_{max} < 1$.

We can rewrite $\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma})$

$$\begin{aligned}\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) &= \Phi(W_n = w_n | W_1 = w_1, W_2 = w_2, \ldots, W_{n-1} = w_{n-1}) \times \\ &\quad \Phi(W_{n-1} = w_{n-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n-2} = w_{n-2}) \times \cdots \times \\ &\quad \Phi(W_2 = w_2 | W_1 = w_1) \times \Phi(W_1 = w_1) \\ &\quad = \Phi(W_1 = w_1) \times \Pi_{k=2}^{n_1+n_2-1} c_k\end{aligned}$$

Then the set

$$\begin{aligned}&\Big\{\Phi(W_{n_1+n_2-1} = w_{n_1+n_2-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\ &\Phi(W_{n_1+n_2-2} = w_{n_1+n_2-2} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \ldots, \\ &\Phi(W_2 = w_2 | W_1 = w_1), \Phi(W_1 = w_1)\Big\} \\ &= \Big\{\Phi(W_1 = w_1), c_2, \ldots, c_{n_1+n_2-1}\Big\}\end{aligned} \tag{5}$$

,

Therefore,

$$\begin{aligned}&\max\Big\{\Phi(W_{n_1+n_2-1} = w_{n_1+n_2-1} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-2} = w_{n_1+n_2-2}), \\ &\Phi(W_{n_1+n_2-2} = w_{n_1+n_2-2} | W_1 = w_1, W_2 = w_2, \ldots, W_{n_1+n_2-3} = w_{n_1+n_2-3}), \ldots, \\ &\Phi(W_2 = w_2 | W_1 = w_1), \Phi(W_1 = w_1)\Big\} \\ &= \max\Big\{\Phi(W_1 = w_1), c_2, \ldots, c_{n_1+n_2-1}\Big\} \\ &= \max\Big\{\Phi(W_1 = w_1), c_{max}\Big\} \\ &:= c\end{aligned} \tag{6}$$

Since all terms in the set above are the cumulative distribution functions of univariate normal distribution and $\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)$, then c is strictly less than 1.

$$\mathbf{\Phi}(\mathbf{W}, \mathbf{\Sigma}) \leq c^{n_1+n_2-1} \tag{7}$$

□

3 Recursion

Proposition 3.1. *Suppose that conditions of Theorem 3.4 where the system inclusion function is* $1_i(X_{i,j} \geq X_{(N_i - n_i + 1)})$, $i = 1, 2$. *Suppose that* $n_2 \geq n_1 > 1$, $N_1 \geq N_2 > 1$, *and fix* $k = j$. *Then condition (5) holds, and the conclusions of Theorem 3.4 follow.*

Proof. For ease of notation we will let $g_{n,i}(x) = I_{i,n-i+1}(x)$. The proof will use two induction arguments. For the base case let $n_1 = n_2$ and $N_1 = N_2$. With these specifications we have that

$$g_{n_1,n_1-j} \circ g^{-1}_{n_2,n_2-j}(x) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(x) = 0.$$

For the induction step, choose some $m \leq n_1 - 1$ and suppose that

$$g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \geq 0, \tag{8}$$

for $t \in (0, 1)$ where $n_2 \geq n_1$ and $N_1 \geq N_2$. An integration by parts calculation applied to regularized incomplete beta functions gives

$$I_{\alpha,\beta}(x) = I_{\alpha+1,\beta}(x) + \frac{x^\alpha (1-x)^\beta}{\alpha B(\alpha, \beta)}.$$

Applying the above to (8) yields

$$\begin{aligned}
&g_{n_1-m-1,n_1-m-1-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\
&\quad = g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\
&\qquad + \frac{\left[g^{-1}_{n_2,n_2-j}(t)\right]^{n_1-m-1-j} \left[1 - g^{-1}_{n_2,n_2-j}(t)\right]^{j+1}}{(n_1 - m - 1 - j) B(n_1 - m - 1 - j, j + 1)} \\
&\quad \geq 0.
\end{aligned}$$

A similar argument holds for iterative increases to N_1. Thus, condition (5) holds under are assumed conditions and the conclusions of Theorem 3.4 follow. □

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then

$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0, 1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha, \beta)$ is the CDF of Beta(α, β) distribution.

4 Proof to Theorem 4.5

Proof. Let $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$ be consistent estimators of the population parameters μ_1, μ_2, σ_1, and σ_2, respectively, as defined in Section 4.2. By replacing the population parameters in $\mathbf{w}$ and $\mathbf{\Sigma}$ with these estimators, we obtain the sample-level analogs $\hat{\mathbf{w}}$ and $\hat{\mathbf{\Sigma}}$. Specifically, we define:

$$\hat{\mathbf{w}} = \left(\frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, \ldots, \frac{\hat{\mu}_2 - \hat{\mu}_1}{\hat{\sigma}_1}, 0, \ldots, 0 \right) \in \mathbb{R}^{n_1+n_2-1},$$

where the first n_1 entries correspond to the standardized mean difference, and the remaining $n_2 - 1$ entries are zero, reflecting the structure introduced in Section 4.2.

The estimated covariance matrix $\hat{\mathbf{\Sigma}}$ has the block form:

$$\hat{\mathbf{\Sigma}} = \begin{bmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{bmatrix},$$

where each submatrix is obtained by replacing the population parameters in $\mathbf{\Sigma}$ with their respective sample estimates. These submatrices retain the same structural forms as defined in Section 4.2.

Based on the Equation (12), we can rewrite

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) \\ &= n_2(\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma)) \end{aligned}$$

Consider the term $\Phi(\hat{\mathbf{w}}, \hat{\mathbf{\Sigma}}) - \Phi(\mathbf{w}, \mathbf{\Sigma})$. We perform a Taylor expansion of this expression with respect to the parameters μ_1, μ_2, σ_1, and σ_2.

$$\begin{aligned} \Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx &\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\ &+ \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2) \end{aligned}$$

Then for the first derivative term $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}$:

$$\begin{aligned} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_1} \\ &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \end{aligned}$$

By applying Leibniz rule and without loss of generality, we compute the $\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_1}$

$$\begin{aligned} \frac{\partial \Phi(\mathbf{w}; \mathbf{\Sigma})}{\partial w_1} &= \frac{\partial}{\partial w_1} \int_{-\infty}^{w_1} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(z_1, \ldots, z_{n_1+n_2-1}) dz_{n_1+n_2-1} \ldots dz_1 \\ &= \int_{-\infty}^{w_2} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{\mathbf{W}}(w_1, z_2, \ldots, z_{n_1+n_2-1}) dz_{n_1+n_2-1} \ldots dz_2 \\ &= \int_{-\infty}^{w_2} \cdots \int_{-\infty}^{w_{n_1+n_2-1}} f_{W_1}(w_1) \cdot f_{\tilde{\mathbf{W}}}(\tilde{w}) dz_{n_1+n_2-1} \ldots dz_2 \\ &= \phi(w_1)\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) \end{aligned}$$

where $\phi(w_1)$ is the probability density function of univariate normal distribution with $\mu = 0$ and $\sigma^2 = \frac{\sigma_2^2}{\sigma_1^2} + 1$, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is the conditional cumulative distribution function of the remaining components given $W_1 = w_1$. By applying Schur complement [Zhang, 2006],

$$\tilde{\mathbf{w}} = (w_2, \ldots, w_{n_1+n_2-1}) - (\frac{\sigma_2^2}{\sigma_1^2}, \ldots, \frac{\sigma_2^2}{\sigma_1^2}, \frac{\sigma_2}{\sigma_1}, \ldots, \frac{\sigma_2}{\sigma_1}) * (\frac{\sigma_2^2}{\sigma_1^2} + 1)^{-1} * (w_1)$$

$\tilde{\mathbf{w}}$ is a vector with the first $n_1 - 1$ elements is $\frac{\mu_2-\mu_1}{\sigma_1} - \frac{\mu_2-\mu_1}{\sigma_1}\frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$ and the last $n_2 - 1$ elements are $-\frac{(\mu_2-\mu_1)\sigma_2}{\sigma_1^2+\sigma_2^2}$.

$$\tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_{-1,-1} - \mathbf{\Sigma}_{-1,1}\mathbf{\Sigma}_{1,1}^{-1}\mathbf{\Sigma}_{1,-1}$$

where $\mathbf{\Sigma}_{-1,-1}$ denotes the submatrix of $\mathbf{\Sigma}$ obtained by removing the first row and the first column, $\mathbf{\Sigma}_{-1,1}$ denotes the column vector obtained by removing the first element from the first column of $\mathbf{\Sigma}$, $\mathbf{\Sigma}_{1,-1}$ denotes the row vector obtained by removing the first element from the first row of $\mathbf{\Sigma}$, and $\mathbf{\Sigma}_{1,1}$ denotes the top-left scalar element of $\mathbf{\Sigma}$.

Let denote the form of the matrix $\tilde{\mathbf{\Sigma}}$:

$$\tilde{\mathbf{\Sigma}} = \begin{bmatrix} \tilde{\Sigma}_{11} & \tilde{\Sigma}_{12} \\ \tilde{\Sigma}_{21} & \tilde{\Sigma}_{22} \end{bmatrix}$$

$$\tilde{\Sigma}_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix}_{(n_1-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^4}{\sigma_1^4} & \cdots & \cdots & \frac{\sigma_2^4}{\sigma_1^4} \end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \frac{\sigma_2^2}{\sigma_1^2}-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \frac{\sigma_2^2}{\sigma_1^2}+1-\frac{\frac{\sigma_2^4}{\sigma_1^4}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_1-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{12}^T = \tilde{\Sigma}_{21} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_1-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^3}{\sigma_1^3} & \cdots & \cdots & \frac{\sigma_2^3}{\sigma_1^3} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} - \frac{\frac{\sigma_2^3}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_1-1)}$$

$$\tilde{\Sigma}_{22} = \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}_{(n_2-1)\times(n_2-1)} - \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

$$= \begin{bmatrix} 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \\ 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & \dots & 1 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} & 2 - \frac{\frac{\sigma_2^2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}+1} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Therefore, $\Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$ is finite. Then for $i = 1, \dots, n_1$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}$ is the same as $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_1}$. By applying Lemma 4.4, we get

$$\begin{aligned} \frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1} &= -\frac{1}{\sigma_1}\sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i} \\ &= -\frac{n_1}{\sigma_1}\cdot\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) \\ &\leq O_P(n_1 c^{n_1+n_2-1}) \end{aligned}$$

By applying the similar method used for $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, we have

$$\begin{aligned} \frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2} &= \sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i}\cdot\frac{\partial w_i}{\partial\mu_2} \\ &= \frac{1}{\sigma_1}\sum_{i=1}^{n_1}\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial w_i} \end{aligned}$$

$$= \frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma})$$
$$= OP(n_1 c^{n_1+n_2-1})$$

Notice that $\Phi(\mathbf{w}, \Sigma)$ has two parameters $\mathbf{w}$, and Σ involving σ_1, then using the Equation (137) in Petersen et al. [2008] and the similar method used for $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}$, we have

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \sigma_1} + \mathrm{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) \\ &= -\frac{(\mu_2 - \mu_1)n_1}{\sigma_1^2} \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}}, \tilde{\Sigma}) + \mathrm{tr}(\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})\end{aligned}$$

Before computing $\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}$, we define the $\phi(\mathbf{x})$ be the probability density function of $\Phi(\mathbf{w}, \Sigma)$, which is

$$\phi(\mathbf{x}) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp\left[-\frac{1}{2}(\mathbf{x})^T \Sigma^{-1}(\mathbf{x})\right]$$

then

$$\log(\phi(\mathbf{x})) = -\frac{1}{2}(\log(\det(2\pi\Sigma)) + \mathbf{x}^t \Sigma^{-1} \mathbf{x})$$

then using the Equation (57) and (61) in Petersen et al. [2008], we have

$$\begin{aligned}\frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} &= -\frac{1}{2}(\frac{\partial \log(\det(2\pi\Sigma))}{\partial \Sigma} + \frac{\partial \mathbf{x}^t \Sigma^{-1} \mathbf{x}}{\partial \Sigma}) \\ &= -\frac{1}{2}(\Sigma^{-1} - \Sigma^{-1} \mathbf{x}\mathbf{x}^t \Sigma^{-1})\end{aligned}$$

Then for the term $\mathrm{tr}(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})$ is:

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \Sigma} &= \frac{\partial \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) d\mathbf{x}}{\partial \Sigma} \\ &= \int_{-\infty}^{\mathbf{w}} \frac{\partial \phi(\mathbf{x})}{\partial \Sigma} d\mathbf{x} \\ &= \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} d\mathbf{x} \\ &= -\frac{1}{2}\Sigma^{-1}\Phi(\mathbf{w}, \Sigma) + \frac{1}{2}\Sigma^{-1} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} \\ &= \frac{1}{2}\Sigma^{-1} \left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} \Sigma^{-1} - \Phi(\mathbf{w}, \Sigma) \right)\end{aligned}$$

Then

$$\operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1}) = \operatorname{tr}\left(\frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)\frac{\partial\Sigma}{\partial\sigma_1}\right)$$
$$= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)$$

To compute the $\operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1})$, we firstly compute the $\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right)$. Let define $\frac{\partial\Sigma}{\partial\sigma_1}$ be:

$$\frac{\partial\Sigma}{\partial\sigma_1} = \begin{bmatrix}\frac{\partial\Sigma_{11}}{\partial\sigma_1} & \frac{\partial\Sigma_{12}}{\partial\sigma_1}\\ \frac{\partial\Sigma_{21}}{\partial\sigma_1} & \frac{\partial\Sigma_{22}}{\partial\sigma_1}\end{bmatrix} = \begin{bmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{bmatrix}$$

$$A_{11} = \left[-\frac{2\sigma_2^2}{\sigma_1^3}\right]_{n_1\times n_1}, A_{12}^T = A_{21} = \left[-\frac{\sigma_2}{\sigma_1^2}\right]_{(n_2-1)\times n_1}, A_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

and by applying Schur complement [Zhang, 2006], we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}^{-1} = \begin{bmatrix}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & -(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}\\ -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & \Sigma_{22}^{-1}+\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}\end{bmatrix}$$

where Σ_{11}, Σ_{12}, Σ_{21}, and Σ_{22} are defined in Equation (10).
By applying Equation (15), we have

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix}B_{11} & B_{12}\\ B_{21} & B_{22}\end{bmatrix}$$

$$B_{11} = (\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$$
$$= \left(\begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix} - \begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\begin{bmatrix}2 & 1 & \dots & 1\\ 1 & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \dots & 1 & 2\end{bmatrix}^{-1}\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\right)^{-1}$$
$$= \left(\begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix} - \begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\begin{bmatrix}\frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2}\\ -\frac{1}{n_2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2}\end{bmatrix}\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}\right)^{-1}$$

$$= \left(\begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1 \end{bmatrix} - (1-\frac{1}{n_2}) \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2^2}{\sigma_1^2} & \cdots & \cdots & \frac{\sigma_2^2}{\sigma_1^2} \end{bmatrix} \right)^{-1}$$

$$= \begin{bmatrix} \frac{\sigma_2^2}{n_2\sigma_1^2}+1 & \frac{\sigma_2^2}{n_2\sigma_1^2} & \cdots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{n_2\sigma_1^2} \\ \frac{\sigma_2^2}{n_2\sigma_1^2} & \cdots & \frac{\sigma_2^2}{n_2\sigma_1^2} & \frac{\sigma_2^2}{n_2\sigma_1^2}+1 \end{bmatrix}^{-1} = \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}_{n_1\times n_1}$$

$$B_{12} = -(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 2 \end{bmatrix}^{-1}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \cdots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix}$$

$$= -\frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \cdots & \cdots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \cdots & \cdots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{n_1\times(n_2-1)}$$

$$B_{21} = -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}$$

$$= -\begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= -\begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix} \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \dots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & -\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix}_{(n_2-1)\times n_1}$$

$$B_{22} = \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1}$$

$$= \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1} \end{bmatrix} \begin{bmatrix} 2 & 1 & \dots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \dots & 1 & 2 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} & \dots & \dots & \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1} \end{bmatrix} \begin{bmatrix} \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{n_2\sigma_1} & \dots & \dots & \frac{\sigma_2}{n_2\sigma_1} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{n_2-1}{n_2} & -\frac{1}{n_2} & \dots & -\frac{1}{n_2} \\ -\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ -\frac{1}{n_2} & \dots & -\frac{1}{n_2} & \frac{n_2-1}{n_2} \end{bmatrix} + \begin{bmatrix} \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \dots & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \dots & \dots & \frac{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}$$

$$= \begin{bmatrix} 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & 1-\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Then we have

$$\Phi(\mathbf{w},\Sigma) \leq c^{n_1+n_2-1}$$

Therefore,

$$\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1} \leq \Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial\Sigma}{\partial\sigma_1}$, we have

$$\begin{aligned}
&\operatorname{tr}(\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}) \\
&= \operatorname{tr}\left(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\right) \\
&= \operatorname{tr}(B_{11}A_{11}+B_{12}A_{21}) + tr(B_{21}A_{12}+B_{22}A_{22}) \\
&= n_1\left(\frac{-\frac{2\sigma_2^2}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}) + \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}(n_2-1)\right) \\
&+ (n_2-1)\left(n_1\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{\sigma_2}{\sigma_1^2}\right) \\
&= -2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}
\end{aligned}$$

Now we have

$$\operatorname{tr}(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}) \leq 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-1}$$

Now we compute the term $\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right)$. Let rewrite the $\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}$ into matrix form:

$$\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}$$

Define $\mathbf{w}_{-i}$ as the vector $\mathbf{w}$ with the ith component removed, and $\mathbf{w}_{-(i,j)}$ as the vector $\mathbf{w}$ with the ith and j th components removed. Similarly, $\mathbf{x}_{-i}$ and $\mathbf{x}_{-(i,j)}$ are defined in the same manner.

Then

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i^2 d\mathbf{x} &= \int_{-\infty}^{w_i} \left(\int_{-\infty}^{\mathbf{w}_{-i}} \phi(\mathbf{x}_{-i}|x_i) d\mathbf{x}_{-i} \right) \; x_i^2 \; \phi(x_i) dx_i \\ &\leq c^{n_1+n_2-3} \cdot Var(X_i) \end{aligned}$$

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i x_j d\mathbf{x} &= \int_{-\infty}^{w_i} \int_{-\infty}^{w_j} \left(\int_{-\infty}^{\mathbf{w}_{-(i,j)}} \phi(\mathbf{x}_{-(i,j)}|x_i, x_j) d\mathbf{x}_{-(i,j)} \right) \; x_i x_j \; \phi(x_i, x_j) dx_i dx_j \\ &\leq c^{n_1+n_2-3} \cdot Cov(X_i, X_j) \end{aligned}$$

Replace $Var(X_i)$ and $Cov(X_i, X_j)$ using the elements of Σ, we have

$$\begin{aligned} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} &= \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix} \\ &\leq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \end{aligned}$$

where

$$S_{11} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & c^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) \end{bmatrix}_{n_1 \times n_1}$$

$$S_{12}^T = S_{21} = \begin{bmatrix} c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & c^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \end{bmatrix}_{(n_2-1) \times n_1}$$

$$S_{22} = \begin{bmatrix} 2c^{n_1+n_2-3} & c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & c^{n_1+n_2-3} \\ c^{n_1+n_2-3} & \dots & c^{n_1+n_2-3} & 2c^{n_1+n_2-3} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Then,

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = c^{n_1+n_2-3} * \Sigma$$

Therefore, followed by the results before,

$$\begin{aligned} &\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})\cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ &\le \operatorname{tr}\left(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ &= \operatorname{tr}\left(\Sigma^{-1}c^{n_1+n_2-3} * \Sigma * \Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ &= c^{n_1+n_2-3}\operatorname{tr}\left(\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ &= 2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-3} \end{aligned}$$

Then

$$\begin{aligned} \operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1}) &= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_1}\right) \\ &= O_p(c^{n_1+n_2-3}) \end{aligned}$$

By applying the similar method used for $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, we have

$$\begin{aligned} \frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1} &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + \operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_1}) \\ &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + O_p(c^{n_1+n_2-3}) \\ &= O_P(n_1c^{n_1+n_2-3}) \end{aligned}$$

Similarly, we compute $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$,

$$\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2} = \operatorname{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_2})$$

$$= \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right) - \frac{1}{2}\operatorname{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_2}\right)$$

To compute $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$, firstly we have

$$\begin{aligned}\frac{\partial\Sigma}{\partial\sigma_2} &= \begin{bmatrix}\frac{\partial\Sigma_{11}}{\partial\sigma_1} & \frac{\partial\Sigma_{12}}{\partial\sigma_1}\\ \frac{\partial\Sigma_{21}}{\partial\sigma_1} & \frac{\partial\Sigma_{22}}{\partial\sigma_1}\end{bmatrix}\\ &= \begin{bmatrix}C_{11} & C_{12}\\ C_{21} & C_{22}\end{bmatrix}\end{aligned}$$

$$C_{11} = \left[\frac{2\sigma_2}{\sigma_1^2}\right]_{n_1\times n_1}, C_{12}^T = C_{21} = \left[\frac{1}{\sigma_1}\right]_{(n_2-1)\times n_1}, C_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

Then by applying Lemma 4.4, we have

$$\Phi(\mathbf{w},\Sigma) \le c^{n_1+n_2-1}$$

Then

$$\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_2} \le \Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}c^{n_1+n_2-1}$$

Based on the structure of the Σ^{-1} and $\frac{\partial\Sigma}{\partial\sigma_2}$, we have

$$\begin{aligned}&\operatorname{tr}(\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2})\\ &= \operatorname{tr}\left(\begin{bmatrix}B_{11} & B_{12}\\ B_{21} & B_{22}\end{bmatrix}\begin{bmatrix}C_{11} & C_{12}\\ C_{21} & C_{22}\end{bmatrix}\right)\\ &= \operatorname{tr}(B_{11}C_{11}+B_{12}C_{21}) + tr(B_{21}C_{12}+B_{22}C_{22})\\ &= n_1\left(\frac{\frac{2\sigma_2}{\sigma_1^2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2}+1-(n_1-1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2})\right) - \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{1}{\sigma_1}n_1(n_2-1)\\ &- (n_2-1)\left(n_1\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\frac{1}{\sigma_1}\right)\\ &= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\end{aligned}$$

Now we have

$$\mathrm{tr}(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_2}) \leq \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-1}$$

$$\begin{aligned}
&\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right) \leq \mathrm{tr}\left(\Sigma^{-1}\begin{bmatrix}S_{11} & S_{12}\\ S_{21} & S_{22}\end{bmatrix}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right)\\
&= c^{n_1+n_2-3}\,\mathrm{tr}\left(\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right)\\
&= \frac{\frac{2\sigma_2}{\sigma_1^2}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}c^{n_1+n_2-3}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2} &= \mathrm{tr}(\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\Sigma}\cdot\frac{\partial\Sigma}{\partial\sigma_2})\\
&= \frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\int_{-\infty}^{\mathbf{w}}\phi(\mathbf{x})\cdot\mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_2}\right) - \frac{1}{2}\,\mathrm{tr}\left(\Sigma^{-1}\Phi(\mathbf{w},\Sigma)\frac{\partial\Sigma}{\partial\sigma_2}\right)\\
&= O_p(c^{n_1+n_2-3})
\end{aligned}$$

Then we put $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2}$, $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1}$, and $\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}$ together and based on the definition of root sample-size consistent estimator,

$$|\hat{\mu}_1-\mu_1| = O_p(n_1^{-1/2}),$$

$$|\hat{\mu}_2-\mu_2| = O_p(n_2^{-1/2}),$$

$$|\hat{\sigma}_1-\sigma_1| = O_p(n_1^{-1/2}),$$

$$|\hat{\sigma}_2-\sigma_2| = O_p(n_2^{-1/2}),$$

we have

$$\begin{aligned}
\Phi(\hat{\mathbf{w}},\hat{\Sigma})-\Phi(\mathbf{w},\Sigma) \approx{}& \frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_1}(\hat{\mu_1}-\mu_1)+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\mu_2}(\hat{\mu_2}-\mu_2)+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_1}(\hat{\sigma_1}-\sigma_1)\\
&+\frac{\partial\Phi(\mathbf{w},\Sigma)}{\partial\sigma_2}(\hat{\sigma_2}-\sigma_2)\\
&= O_P(n_1^{\frac{1}{2}}c^{n_1+n_2-3})
\end{aligned}$$

Finally,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}|\hat{\mu_1}.\hat{\mu_2},\hat{\sigma_1},\hat{\sigma_2}\right)-\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}|\mu_1.\mu_2,\sigma_1,\sigma_2\right)\\
&= O_P(n_1^{\frac{1}{2}}n_2c^{n_1+n_2-3}+n_1n_2^{\frac{1}{2}}c^{n_1+n_2-3})
\end{aligned}$$

□

5 Simulation Results for Theorem 4.3

We now show the simulation results of our Normal Distribution example under different configurations of $(\mu_1, \mu_2, \sigma_1, \sigma_2, n_1, n_2, N_1, N_2)$. We verify Theorem 4.3 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 2000. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (1) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The theoretical results are computed from the Theorem 4.3 using mvtnorm package [Genz and Bretz, 2009] in R

The results are tabulated in Table 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

n_1	n_2	N_1	N_2	μ_1	μ_2	σ_1	σ_2	Simulation Prob	Theoretical Prob
10	40	400	600	-1	1	1.0	3	0.4000	0.4000099
10	20	400	600	0	0	1.0	3	0.3970	0.3970305
10	40	400	300	-1	1	1.0	3	0.5715	0.5714174
10	20	400	600	0	1	1.5	2	0.3645	0.3647783
10	40	400	600	0	1	1.0	3	0.3995	0.3999245
20	40	400	600	0	1	1.0	3	0.3995	0.3999803
10	40	400	600	-1	0	1.5	3	0.4000	0.3995149
20	40	400	600	0	0	1.5	2	0.2970	0.2962431
20	40	400	600	0	0	1.0	3	0.3750	0.3740938
10	20	400	600	-1	1	1.0	3	0.3990	0.4000059

Table 1: Simulation Results for Normal Example.

References

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